

Chapter 8: Continuity

EXERCISE 8.1 [PAGES 111 - 112]

Exercise 8.1 | Q 1.1 | Page 111

Examine the continuity of $f(x) = x^3 + 2x^2 - x - 2$ at $x = -2$.

SOLUTION

$$f(x) = x^3 + 2x^2 - x - 2$$

Here $f(x)$ is a polynomial function and hence

it is continuous for all $x \in \mathbb{R}$.

$\therefore f(x)$ is continuous at $x = -2$.

Exercise 8.1 | Q 1.2 | Page 111

Examine the continuity of $f(x) = \frac{x^2 - 9}{x - 3}$ on $\mathbb{R}$.

SOLUTION

$$f(x) = \frac{x^2 - 9}{x - 3}; x \in \mathbb{R}$$

$f(x)$ is a rational function and is continuous for all $x \in \mathbb{R}$, except at the points where denominator becomes zero. Here, denominator $x - 3 = 0$ when $x = 3$.

$\therefore$ Function f is continuous for all $x \in \mathbb{R}$, except at $x = 3$, where it is not defined.

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Examine whether the function is continuous at the points indicated against them.

$$f(x) = \begin{cases} x^3 - 2x + 1, & \text{for } x \leq 2 \\ 3x - 2, & \text{for } x > 2, \text{ at } x = 2. \end{cases}$$

SOLUTION

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (x^3 - 2x + 1)$$

$$= (2)^3 - 2(2) + 1 = 5$$

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (3x - 2)$$

$$= 3(2) - 2 = 4$$

$$\therefore \lim_{x \rightarrow 2^-} f(x) \neq \lim_{x \rightarrow 2^+} f(x)$$

$\therefore$ Function f is discontinuous at $x = 2$

Exercise 8.1 | Q 2.2 | Page 111

Examine whether the function is continuous at the points indicated against them.

$$f(x) = \frac{x^2 + 18x - 19}{x - 1} \quad \text{for } x \neq 1$$

$$= 20 \quad \text{for } x = 1, \text{ at } x = 1$$

SOLUTION

$$\begin{aligned} \lim_{x \rightarrow 1} f(x) &= \lim_{x \rightarrow 1} \frac{x^2 + 18x - 19}{x - 1} \\ &= \lim_{x \rightarrow 1} \frac{x^2 + 19x - x - 19}{x - 1} \\ &= \lim_{x \rightarrow 1} \frac{x(x + 19) - 1(x + 19)}{x - 1} \\ &= \lim_{x \rightarrow 1} \frac{(x - 1)(x + 19)}{(x - 1)} \\ &= \lim_{x \rightarrow 1} (x + 19) \quad \dots [\because x \rightarrow 1, \therefore x \neq 1, \therefore x - 1 \neq 0] \\ &= 1 + 19 = 20 \end{aligned}$$

$$\text{Also, } f(1) = 20$$

$$\therefore \lim_{x \rightarrow 1} f(x) = f(1)$$

$\therefore f(x)$ is continuous at $x = 1$

Test the continuity of the following function at the points indicated against them.

$$f(x) = \frac{\sqrt{x-1} - (x-1)^{\frac{1}{3}}}{x-2} \quad \text{for } x \neq 2$$

$$= \frac{1}{5} \quad \text{for } x = 2, \text{ at } x = 2$$

SOLUTION

$$f(2) = \frac{1}{5} \quad \dots\dots(\text{given})$$

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} \frac{\sqrt{x-1} - (x-1)^{\frac{1}{3}}}{x-2}$$

$$\text{Put } x - 1 = y$$

$$\therefore x = 1 + y$$

$$\therefore \text{As } x \rightarrow 2, y \rightarrow 1$$

$$\therefore \lim_{x \rightarrow 2} f(x) = \lim_{y \rightarrow 1} \frac{\sqrt{y} - y^{\frac{1}{3}}}{1 + y - 2}$$

$$= \lim_{y \rightarrow 1} \frac{y^{\frac{1}{2}} - 1 - y^{\frac{1}{3}} + 1}{y - 1}$$

$$= \lim_{y \rightarrow 1} \frac{\left(y^{\frac{1}{2}} - 1\right) - \left(y^{\frac{1}{3}} - 1\right)}{y - 1}$$

$$= \lim_{y \rightarrow 1} \left(\frac{y^{\frac{1}{2}} - 1}{y - 1} - \frac{y^{\frac{1}{3}} - 1}{y - 1} \right)$$

$$\begin{aligned}
&= \lim_{y \rightarrow 1} \frac{y^{\frac{1}{2}} - 1^{\frac{1}{2}}}{y - 1} - \lim_{y \rightarrow 1} \frac{y^{\frac{1}{3}} - 1^{\frac{1}{3}}}{y - 1} \\
&= \frac{1}{2} (1)^{-\frac{1}{2}} - \frac{1}{3} (1)^{-\frac{2}{3}} \quad \dots\dots [\because \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = n \cdot a^{n-1}] \\
&= \frac{1}{2} - \frac{1}{3} \\
&= \frac{1}{6}
\end{aligned}$$

$$\therefore \lim_{x \rightarrow 2} f(x) \neq f(2)$$

$\therefore f(x)$ is discontinuous at $x = 2$

Exercise 8.1 | Q 3.2 | Page 112

Test the continuity of the following function at the points indicated against them.

$$\begin{aligned}
f(x) &= \frac{x^3 - 8}{\sqrt{x+2} - \sqrt{3x-2}} \quad \text{for } x \neq 2 \\
&= -24 \quad \text{for } x = 2, \text{ at } x = 2
\end{aligned}$$

SOLUTION

$$f(2) = -24 \quad \dots\dots\dots (\text{given})$$

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} \frac{x^3 - 8}{\sqrt{x+2} - \sqrt{3x-2}}$$

$$\begin{aligned}
&= \lim_{x \rightarrow 2} \frac{x^3 - 8}{\sqrt{x+2} - \sqrt{3x-2}} \times \frac{\sqrt{x+2} + \sqrt{3x-2}}{\sqrt{x+2} + \sqrt{3x-2}} \\
&= \lim_{x \rightarrow 2} \frac{(x^3 - 8)(\sqrt{x+2} + \sqrt{3x-2})}{(x+2) - (3x-2)} \\
&= \lim_{x \rightarrow 2} \frac{(x^3 - 2^3)(\sqrt{x+2} + \sqrt{3x-2})}{-2x + 4} \\
&= \lim_{x \rightarrow 2} \frac{(x-2)(x^2 + 2x + 4)(\sqrt{x+2} + \sqrt{3x-2})}{-2(x-2)} \\
&= \lim_{x \rightarrow 2} \frac{(x^2 + 2x + 4)(\sqrt{x+2} + \sqrt{3x-2})}{-2} \quad \text{.....} [\because x \rightarrow 2, x \neq 2 \therefore x-2 \neq 0] \\
&= \frac{-1}{2} \lim_{x \rightarrow 2} (x^2 + 2x + 4)(\sqrt{x+2} + \sqrt{3x-2}) \\
&= \frac{-1}{2} \lim_{x \rightarrow 2} (x^2 + 2x + 4) \lim_{x \rightarrow 2} (\sqrt{x+2} + \sqrt{3x-2}) \\
&= \frac{-1}{2} \times [2^2 + 2(2) + 4] \times (\sqrt{2+2} + \sqrt{3(2)-2}) \\
&= \frac{-1}{2} \times 12 \times (2+2) \\
&= -24
\end{aligned}$$

$$\therefore \lim_{x \rightarrow 2} f(x) = f(2)$$

$\therefore f(x)$ is continuous at $x = 2$

Exercise 8.1 | Q 3.3 | Page 112

Test the continuity of the following function at the points indicated against them.

$$\begin{aligned}
f(x) &= 4x + 1, & \text{for } x \leq 3 \\
&= \frac{59 - 9x}{3}, & \text{for } x > 3 \text{ at } x = \frac{8}{3}.
\end{aligned}$$

SOLUTION

$$\lim_{x \rightarrow (\frac{8}{3})^-} f(x) = \lim_{x \rightarrow (\frac{8}{3})^-} (4x + 1)$$

$$= 4\left(\frac{8}{3}\right) + 1$$

$$= \frac{32}{3} + 1$$

$$= \frac{35}{3}$$

$$\lim_{x \rightarrow (\frac{8}{3})^+} f(x) = \lim_{x \rightarrow (\frac{8}{3})^+} \frac{59 - 9x}{3}$$

$$= \frac{59 - 9\left(\frac{8}{3}\right)}{3}$$

$$= \frac{59 - 24}{3}$$

$$= \frac{35}{3}$$

$$f(x) = 4x + 1, \quad x \leq \left(\frac{8}{3}\right)$$

$$\therefore f\left(\frac{8}{3}\right) = 4\left(\frac{8}{3}\right) + 1$$

$$= \frac{32}{3} + 1$$

$$= \frac{35}{3}$$

$$\lim_{x \rightarrow (\frac{8}{3})^-} f(x) = \lim_{x \rightarrow (\frac{8}{3})^+} f(x) = f\left(\frac{8}{3}\right)$$

$$\therefore f(x) \text{ is continuous at } x = \frac{8}{3}$$

Test the continuity of the following function at the points indicated against them.

$$f(x) = \frac{x^3 - 27}{x^2 - 9} \text{ for } 0 \leq x < 3$$

$$= \frac{9}{2} \text{ for } 3 \leq x \leq 6$$

at $x = 3$

SOLUTION

$$f(3) = \frac{9}{2} \text{(given)}$$

$$\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} \frac{x^3 - 27}{x^2 - 9}$$

$$= \lim_{x \rightarrow 3} \frac{(x - 3)(x^2 + 3x + 9)}{(x - 3)(x + 3)}$$

$$= \lim_{x \rightarrow 3} \frac{x^2 + 3x + 9}{x + 3} \quad [\text{As } x \rightarrow 3, x \neq 3 \therefore x - 3 \neq 0]$$

$$= \frac{(3)^2 + 3(3) + 9}{3 + 3}$$

$$= \frac{9 + 9 + 9}{6}$$

$$= \frac{27}{6}$$

$$= \frac{9}{2}$$

$$\therefore \lim_{x \rightarrow 3} f(x) = f(3)$$

$\therefore$ Function f is continuous at $x = 3$

Exercise 8.1 | Q 4.1 | Page 112

If $f(x) = \frac{24^x - 8^x - 3^x + 1}{12^x - 4^x - 3^x + 1}$ for $x \neq 0$

$= k,$ for $x = 0$ is continuous at $x = 0$, find k .

SOLUTION

Function f is continuous at $x = 0$

$$\therefore f(0) = \lim_{x \rightarrow 0} f(x)$$

$$\begin{aligned} \therefore k &= \lim_{x \rightarrow 0} \frac{24^x - 8^x - 3^x + 1}{12^x - 4^x - 3^x + 1} \\ &= \lim_{x \rightarrow 0} \frac{8^x \cdot 3^x - 8^x - 3^x + 1}{4^x \cdot 3^x - 4^x - 3^x + 1} \\ &= \lim_{x \rightarrow 0} \frac{8^x(3^x - 1) - 1(3^x - 1)}{4^x(3^x - 1) - 1(3^x - 1)} \\ &= \lim_{x \rightarrow 0} \frac{(3^x - 1)(8^x - 1)}{(3^x - 1)(4^x - 1)} \\ &= \lim_{x \rightarrow 0} \frac{8^x - 1}{4^x - 1} \quad [[\because x \rightarrow 0, 3^x \rightarrow 3^0], [\because 3^x \rightarrow 1 \therefore 3^x \neq 1], [\because 3^x - 1 \neq 0]] \\ &= \lim_{x \rightarrow 0} \left(\frac{\frac{8^x - 1}{x}}{\frac{4^x - 1}{x}} \right) \quad \dots [\because x \rightarrow 0, \therefore x \neq 0] \\ &= \frac{\log 8}{\log 4} \quad \dots \left[\because \lim_{x \rightarrow 0} \left(\frac{a^x - 1}{x} \right) = \log a \right] \\ &= \frac{\log(2)^3}{\log(2)^2} \\ &= \frac{3 \log 2}{3 \log 2} \\ \therefore f(0) &= \frac{3}{2} \end{aligned}$$

Exercise 8.1 | Q 4.2 | Page 112

$$\text{If } f(x) = \frac{5^x + 5^{-x} - 2}{x^2} \quad \text{for } x \neq 0$$

$= k$ for $x = 0$ is continuous at $x = 0$, find k

SOLUTION

Function f is continuous at $x = 0$

$$\therefore f(0) = \lim_{x \rightarrow 0} f(x)$$

$$\therefore k = \lim_{x \rightarrow 0} \frac{5^x + 5^{-x} - 2}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{5^x + \frac{1}{5^x} - 2}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{(5^x)^2 + 1 - 2(5^x)}{5^x \cdot x^2}$$

$$= \lim_{x \rightarrow 0} \frac{(5^x - 1)^2}{5^x \cdot x^2} \dots [\because a^2 - 2ab + b^2 = (a - b)^2]$$

$$= \lim_{x \rightarrow 0} \left(\frac{5^x - 1}{x} \right)^2 \cdot \frac{1}{5^x}$$

$$= \lim_{x \rightarrow 0} \left(\frac{5^x - 1}{x} \right)^2 \times \lim_{x \rightarrow 0} \frac{1}{5^x}$$

$$= (\log 5)^2 \times \frac{1}{5^0} \dots \left[\because \lim_{x \rightarrow 0} \left(\frac{a^x - 1}{x} \right) = \log a \right]$$

$$\therefore k = (\log 5)^2$$

Exercise 8.1 | Q 4.3 | Page 112

For what values of a and b is the function

$$\begin{aligned} f(x) &= ax + 2b + 18 && \text{for } x \leq 0 \\ &= x^2 + 3a - b && \text{for } 0 < x \leq 2 \\ &= 8x - 2 && \text{for } x > 2, \end{aligned}$$

continuous for every x ?

SOLUTION

Function f is continuous for every x.

$\therefore$ Function f is continuous at $x = 0$ and $x = 2$

As f is continuous at $x = 0$.

$$\therefore \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x)$$

$$\therefore \lim_{x \rightarrow 0^-} (ax + 2b + 18) = \lim_{x \rightarrow 0^+} (x^2 + 3a - b)$$

$$\therefore a(0) + 2b + 18 = (0)^2 + 3a - b$$

$$\therefore 3a - 3b = 18$$

$$\therefore a - b = 6 \quad \dots\dots\dots(i)$$

Also, Function f is continuous at $x = 2$

$$\therefore \lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x)$$

$$\therefore \lim_{x \rightarrow 2^-} (x^2 + 3a - b) = \lim_{x \rightarrow 2^+} (8x - 2)$$

$$\therefore (2)^2 + 3a - b = 8(2) - 2$$

$$\therefore 4 + 3a - b = 14$$

$$\therefore 3a - b = 10 \quad \dots(ii)$$

Subtracting (i) from (ii), we get

$$2a = 4$$

$$\therefore a = 2$$

Substituting $a = 2$ in (i), we get

$$2 - b = 6$$

$$\therefore b = -4$$

$$\therefore a = 2 \text{ and } b = -4$$

For what values of a and b is the function

$$f(x) = \frac{x^2 - 4}{x - 2} \quad \text{for } x < 2$$

$$= ax^2 - bx + 3 \quad \text{for } 2 \leq x < 3$$

$$= 2x - a + b \quad \text{for } x \geq 3$$

continuous in its domain.

SOLUTION

Function f is continuous for every x on R.

∴ Function f is continuous at x = 2 and x = 3.

As f is continuous at x = 2.

$$\therefore \lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x)$$

$$\therefore \lim_{x \rightarrow 2^-} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2^+} (ax^2 - bx + 3)$$

$$\therefore \lim_{x \rightarrow 2^-} \frac{(x - 2)(x + 2)}{x - 2} = \lim_{x \rightarrow 2^+} (ax^2 - bx + 3)$$

$$\therefore \lim_{x \rightarrow 2^-} (x + 2) = \lim_{x \rightarrow 2^+} (ax^2 - bx + 3) \dots\dots [\because x \rightarrow 2 \therefore x \neq 2 \therefore x - 2 \neq 0]$$

$$\therefore 2 + 2 = a(2)^2 - b(2) + 3$$

$$\therefore 4 = 4a - 2b + 3$$

$$\therefore 4a - 2b = 1 \dots(i)$$

Also function f is continuous at x = 3

$$\therefore \lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x)$$

$$\therefore \lim_{x \rightarrow 3^-} (ax^2 - bx + 3) = \lim_{x \rightarrow 3^+} (2x - a + b)$$

$$\therefore a(3)^2 - b(3) + 3 = 2(3) - a + b$$

$$\therefore 9a - 3b + 3 = 6 - a + b$$

$$\therefore 10a - 4b = 3 \dots(ii)$$

Multiplying (i) by 2, we get

$$8a - 4b = 2 \dots(iii)$$

Subtracting (iii) from (ii), we get

$$2a = 1$$

$$\therefore a = \frac{1}{2}$$

Substituting $a = \frac{1}{2}$ in (i), we get

$$4\left(\frac{1}{2}\right) - 2b = 1$$

$$\therefore 2 - 2b = 1$$

$$\therefore 1 = 2b$$

$$\therefore b = \frac{1}{2}$$

$$\therefore a = \frac{1}{2} \text{ and } b = \frac{1}{2}$$

MISCELLANEOUS EXERCISE 8 [PAGE 113]

Miscellaneous Exercise 8 | Q 1.1 | Page 113

Discuss the continuity of the following function at the point(s) or in the interval indicated against them.

$$f(x) = 2x^2 - 2x + 5 \text{ for } 0 \leq x < 2$$

$$= \frac{1 - 3x - x^2}{1 - x} \text{ for } 2 \leq x < 4$$

$$= \frac{7 - x^2}{x - 5} \text{ for } 4 \leq x \leq 7 \text{ on its domain.}$$

Miscellaneous Exercise 8 | Q 1.2 | Page 113

Discuss the continuity of the following function at the point(s) or in the interval indicated against them.

$$f(x) = \frac{3^x + 3^{-x} - 2}{x^2} \text{ for } x \neq 0$$

$$= (\log 3)^2 \text{ for } x = 0, \text{ at } x = 0$$

SOLUTION

$$f(0) = (\log 3)^2 \dots (\text{given})$$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{3^x + 3^{-x} - 2}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{3^x + \frac{1}{3^x} - 2}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{(3^x)^2 + 1 - 2(3^x)}{x^2 \cdot (3)^x}$$

$$= \lim_{x \rightarrow 0} \frac{(3^x - 1)^2}{x^2 \cdot (3)^x} \dots [\because a^2 - 2ab + b^2 = (a - b)^2]$$

$$= \lim_{x \rightarrow 0} \left[\left(\frac{3^x - 1}{x} \right)^2 \times \frac{1}{3^x} \right]$$

$$= \lim_{x \rightarrow 0} \left(\frac{3^x - 1}{x} \right)^2 \times \frac{1}{\lim_{x \rightarrow 0} 3^x}$$

$$= (\log 3)^2 \times \frac{1}{3^0} \dots \left[\because \lim_{x \rightarrow 0} \left(\frac{a^x - 1}{x} \right) = \log a \right]$$

$$= (\log 3)^2 \times \frac{1}{1}$$

$$= (\log 3)^2$$

$$\therefore \lim_{x \rightarrow 0} f(x) = f(0)$$

$\therefore f$ is continuous at $x = 0$

Miscellaneous Exercise 8 | Q 1.3 | Page 113

Discuss the continuity of the following function at the point(s) or in the interval indicated against them.

$$f(x) = \frac{5^x - e^x}{2x} \text{ for } x \neq 0$$

$$= \frac{1}{2}(\log 5 - 1) \text{ for } x = 0 \text{ at } x = 0$$

SOLUTION

$$f(0) = \frac{1}{2}(\log 5 - 1) \text{ ...[given]}$$

$$\begin{aligned} \lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} \frac{5^x - e^x}{2x} \\ &= \lim_{x \rightarrow 0} \frac{5^x - 1 - e^x + 1}{2x} \\ &= \frac{1}{2} \lim_{x \rightarrow 0} \frac{(5^x - 1) - (e^x - 1)}{x} \\ &= \frac{1}{2} \lim_{x \rightarrow 0} \left[\frac{(5^x - 1)}{x} - \frac{(e^x - 1)}{x} \right] \\ &= \frac{1}{2} \left(\lim_{x \rightarrow 0} \frac{5^x - 1}{x} - \lim_{x \rightarrow 0} \frac{e^x - 1}{x} \right) \\ &= \frac{1}{2} (\log 5 - \log e) \text{ } \left[\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a \right] \\ &= \frac{1}{2} (\log 5 - 1) \text{ ...} [\because \log e = 1] \end{aligned}$$

$$\therefore \lim_{x \rightarrow 0} f(x) = f(0)$$

$\therefore f$ is continuous at $x = 0$

Miscellaneous Exercise 8 | Q 1.4 | Page 113

$$f(x) = \frac{\sqrt{x+3} - 2}{x^3 - 1} \quad \text{for } x \neq 1$$

$$= 2 \quad \text{for } x = 1, \text{ at } x = 1.$$

SOLUTION

$$f(1) = 2 \quad \dots [\text{given}]$$

$$\begin{aligned} \lim_{x \rightarrow 1} f(x) &= \lim_{x \rightarrow 1} \frac{\sqrt{x+3} - 2}{x^3 - 1} \\ &= \lim_{x \rightarrow 1} \left(\frac{\sqrt{x+3} - 2}{x^3 - 1} \times \frac{\sqrt{x+3} + 2}{\sqrt{x+3} + 2} \right) \\ &= \lim_{x \rightarrow 1} \left(\frac{x + 3 - 4}{(x^3 - 1)(\sqrt{x+3} + 2)} \right) \\ &= \lim_{x \rightarrow 1} \frac{x - 1}{(x - 1)(x^2 + x + 1)(\sqrt{x+3} + 2)} \\ &= \lim_{x \rightarrow 1} \frac{1}{(x^2 + x + 1)(\sqrt{x+3} + 2)} \quad \dots [\text{As } x \rightarrow 1, x \neq 1 \therefore x - 1 \neq 0] \\ &= \frac{1}{\lim_{x \rightarrow 1} (x^2 + x + 1) \times \lim_{x \rightarrow 1} (\sqrt{x+3} + 2)} \\ &= \frac{1}{(1^2 + 1 + 1) \times (\sqrt{1+3} + 2)} \\ &= \frac{1}{3}(2 + 2) \end{aligned}$$

$$= \frac{1}{12}$$

$$\therefore \lim_{x \rightarrow 1} f(x) \neq f(1)$$

$\therefore f$ is discontinuous at $x = 1$

Miscellaneous Exercise 8 | Q 1.5 | Page 113

$$f(x) = \frac{\log x - \log 3}{x - 3} \text{ for } x \neq 3$$

$$= 3 \quad \text{for } x = 3, \text{ at } x = 3.$$

SOLUTION

$$f(3) = 3 \dots [\text{given}]$$

$$\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} \frac{\log x - \log 3}{x - 3}$$

Substitute $x - 3 = h$

$$\therefore x = 3 + h,$$

as $x \rightarrow 3$, $h \rightarrow 0$

$$\therefore \lim_{x \rightarrow 3} f(x) = \lim_{h \rightarrow 0} \frac{\log(h + 3) - \log 3}{3 + h - 3}$$

$$= \lim_{h \rightarrow 0} \frac{\log\left(\frac{h+3}{3}\right)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\log\left(1 + \frac{h}{3}\right)}{\left(\frac{h}{3}\right)} \times \frac{1}{3}$$

$$= \frac{1}{3} \lim_{h \rightarrow 0} \frac{\log\left(1 + \frac{h}{3}\right)}{\left(\frac{h}{3}\right)}$$

$$= \frac{1}{3} (1) \dots \left[\because \lim_{x \rightarrow 0} \frac{\log(1 + x)}{x} = 1 \right]$$

$$= \frac{1}{3}$$

$$\therefore \lim_{x \rightarrow 3} f(x) \neq f(3)$$

$\therefore f$ is discontinuous at $x = 3$

Miscellaneous Exercise 8 | Q 2.1 | Page 113

Find k if the following function is continuous at the points indicated against them.

$$f(x) = \left(\frac{5x - 8}{8 - 3x} \right)^{\frac{3}{2x-4}} \text{ for } x \neq 2$$

$$= k \quad \text{for } x = 2 \text{ at } x = 2.$$

SOLUTION

f is continuous at $x = 2$

$$\therefore f(2) = \lim_{x \rightarrow 2} f(x)$$

$$\therefore k = \lim_{x \rightarrow 2} \left(\frac{5x - 8}{8 - 3x} \right)^{\frac{3}{2x-4}}$$

Substitute $x - 2 = h$

$$\therefore x = 2 + h$$

As $x \rightarrow 2$, $h \rightarrow 0$

$$\therefore k = \lim_{h \rightarrow 0} \left[\frac{5(2 + h) - 8}{8 - 3(2 + h)} \right]^{\frac{3}{2(2+h)-4}}$$

$$= \lim_{h \rightarrow 0} \left(\frac{10 + 5h - 8}{8 - 6 - 3h} \right)^{\frac{3}{2h}}$$

$$= \lim_{h \rightarrow 0} \left(\frac{2 + 5h}{2 - 3h} \right)^{\frac{3}{2h}}$$

$$\begin{aligned}
&= \lim_{h \rightarrow 0} \frac{\left(1 + \frac{5h}{2}\right)^{\frac{3}{2h}}}{\left(1 - \frac{3h}{2}\right)^{\frac{3}{2h}}} \\
&= \frac{\lim_{h \rightarrow 0} \left[\left(1 + \frac{5h}{2}\right)^{\frac{2}{5h}}\right]^{\frac{5}{2} \times \frac{3}{2}}}{\lim_{h \rightarrow 0} \left[\left(1 - \frac{3h}{2}\right)^{\frac{2}{3h}}\right]^{\frac{-3}{2} \times \frac{3}{2}}} \\
&= \frac{e^{\frac{15}{4}}}{e^{\frac{-9}{4}}} \dots \left[\because h \rightarrow 0, \frac{5h}{2} \rightarrow 0, \frac{-3h}{2} \rightarrow 0 \text{ and } \lim_{x \rightarrow 0} (1 + x)^{\frac{1}{x}} = e \right] \\
&= e^{\frac{24}{4}}
\end{aligned}$$

$$\therefore k = e^6$$

Miscellaneous Exercise 8 | Q 2.2 | Page 113

Find k if the following function is continuous at the points indicated against them.

$$\begin{aligned}
f(x) &= \frac{45^x - 9^x - 5^x + 1}{(k^x - 1)(3^x - 1)} \text{ for } x \neq 0 \\
&= \frac{2}{3} \text{ for } x = 0, \text{ at } x = 0
\end{aligned}$$

SOLUTION

f is continuous at $x = 0$

$$\therefore \lim_{x \rightarrow 0} f(x) = f(0)$$

$$\therefore \lim_{x \rightarrow 0} \frac{(45)^x - 9^x - 5^x + 1}{(k^x - 1)(3^x - 1)} = \frac{2}{3}$$

$$\therefore \lim_{x \rightarrow 0} \frac{9^x \cdot 5^x - 9^x - 5^x + 1}{(k^x - 1)(3^x - 1)} = \frac{2}{3}$$

$$\therefore \lim_{x \rightarrow 0} \frac{9^x(5^x - 1) - 1(5^x - 1)}{(k^x - 1)(3^x - 1)} = \frac{2}{3}$$

$$\therefore \lim_{x \rightarrow 0} \frac{(5^x - 1)(9^x - 1)}{(k^x - 1)(3^x - 1)} = \frac{2}{3}$$

$$\therefore \lim_{x \rightarrow 0} \frac{\frac{(5^x - 1)(9^x - 1)}{x^2}}{\frac{(k^x - 1)(3^x - 1)}{x^2}} = \frac{2}{3} \quad \dots [\because x \rightarrow 0, \therefore x \neq 0 \therefore x^2 \neq 0 \text{ Divide Numerator and Denominator by } x^2]$$

$$\therefore \frac{\lim_{x \rightarrow 0} \left(\frac{5^x - 1}{x} \right) \left(\frac{9^x - 1}{x} \right)}{\lim_{x \rightarrow 0} \left(\frac{k^x - 1}{x} \right) \left(\frac{3^x - 1}{x} \right)} = \frac{2}{3}$$

$$\therefore \frac{\log 5 \cdot \log 9}{\log k \cdot \log 3} = \frac{2}{3} \quad \left[\because \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a \right]$$

$$\therefore \frac{\log 5 \cdot \log(3)^2}{\log k \cdot \log 3} = \frac{2}{3}$$

$$\therefore \frac{2 \times \log 5 \times \log 3}{\log k \times \log 3} = \frac{2}{3}$$

$$\therefore \frac{\log 5}{\log k} = \frac{1}{3}$$

$$\therefore 3 \log 5 = \log k$$

$$\therefore \log(5)^3 = \log k$$

$$\therefore (5)^3 = k$$

$$\therefore k = 125$$

Miscellaneous Exercise 8 | Q 2.3 | Page 113

Find k if the following function is continuous at the points indicated against them.

$$f(x) = (1 + kx)^{\frac{1}{x}}, \text{ for } x \neq 0$$

$$= e^{\frac{3}{2}}, \text{ for } x = 0, \text{ at } x = 0$$

SOLUTION

f is continuous at x = 0

$$\therefore \lim_{x \rightarrow 0} f(x) = f(0)$$

$$\therefore \lim_{x \rightarrow 0} (1 + kx)^{\frac{1}{x}} = e^{\frac{3}{2}}$$

$$\therefore \lim_{x \rightarrow 0} \left[(1 + kx)^{\frac{1}{kx}} \right]^k = e^{\frac{3}{2}}$$

$$\therefore e^k = e^{\frac{3}{2}} \dots \left[\lim_{x \rightarrow 0} (1 + x)^{\frac{1}{x}} = e \right]$$

$$\therefore k = \frac{3}{2}$$

Miscellaneous Exercise 8 | Q 3.1 | Page 113

Find a and b if the following function is continuous at the point indicated against them.

$$f(x) = x^2 + a, \text{ for } x \geq 0$$

$$= 2\sqrt{x^2 + 1} + b, \text{ for } x < 0 \text{ and } f(1) = 2 \text{ is continuous at } x = 0$$

SOLUTION

Since, $f(x) = x^2 + a, \quad x \geq 0$

$$\therefore f(1) = (1)^2 + a$$

$$\therefore 2 = 1 + a \dots\dots\dots[\because f(1) = 2]$$

$$\therefore a = 1$$

Also f is continuous at $x = 0$

$$\therefore \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x)$$

$$\therefore \lim_{x \rightarrow 0^-} \left(2\sqrt{x^2 + 1} + b \right) = \lim_{x \rightarrow 0^+} (x^2 + a)$$

$$\therefore \left(2\sqrt{0^2 + 1} + b \right) = 0^2 + 1$$

$$\therefore 2\sqrt{0^2 + 1} + b = 0^2 + 1$$

$$\therefore 2(1) + b = 1$$

$$\therefore b = -1$$

$$\therefore a = 1 \text{ and } b = -1$$

Miscellaneous Exercise 8 | Q 3.2 | Page 113

Find a and b if the following function is continuous at the point indicated against them.

$$f(x) = \frac{x^2 - 9}{x - 3} + a, \text{ for } x > 3$$

$$= 5, x = 3$$

$$= 2x^2 + 3x + b, \text{ for } x < 3$$

is continuous at $x = 3$

SOLUTION

f is continuous at $x = 3$

$$\therefore f(3) = \lim_{x \rightarrow 3^-} f(x)$$

$$= \lim_{x \rightarrow 3^-} (2x^2 + 3x + b)$$

$$\therefore 5 = 2(3)^2 + 3(3) + b$$

$$\therefore 5 = 18 + 9 + b$$

$$\therefore b = -22$$

$$\text{Also, } f(3) = f(3) = \lim_{x \rightarrow 3^+} f(x)$$

$$\therefore 5 = \lim_{x \rightarrow 3^+} \frac{x^2 - 9}{x - 3} + a$$

$$= \lim_{x \rightarrow 3^+} \frac{(x + 3)(x - 3)}{(x - 3)} + a$$

$$= \lim_{x \rightarrow 3^+} (x + 3) + a \dots\dots[\because x \rightarrow 3; x \neq 3 \therefore x - 3 \neq 0]$$

$$= (3 + 3) + a$$

$$\therefore 5 = 6 + a$$

$$\therefore a = -1$$

$$\therefore a = -1, b = -22$$

Miscellaneous Exercise 8 | Q 3.3 | Page 113

Find a and b if the following function is continuous at the point indicated against them.

$$f(x) = \frac{32^x - 1}{8^x - 1} + a, \text{ for } x > 0$$

$$= 2, \text{ for } x = 0$$

$$= x + 5 - 2b, \text{ for } x < 0$$

is continuous at $x = 0$

SOLUTION

f is continuous at $x = 0$

$$\therefore \lim_{x \rightarrow 0^-} f(x) = f(0)$$

$$\therefore \lim_{x \rightarrow 0^-} (x + 5 - 2b) = 2$$

$$\therefore 0 + 5 - 2b = 2$$

$$\therefore 5 - 2 = 2b$$

$$\therefore 2b = 3$$

$$\therefore b = \frac{3}{2}$$

$$\text{Also } \lim_{x \rightarrow 0^+} f(x) = f(0)$$

$$\therefore \lim_{x \rightarrow 0^+} \left(\frac{32^x - 1}{8^x - 1} + a \right) = 2$$

$$\therefore \lim_{x \rightarrow 0^+} \left(\frac{\frac{32^x - 1}{x}}{\frac{8^x - 1}{x}} \right) + \lim_{x \rightarrow 0^+} a = 2$$

$$\therefore \frac{\log 32}{\log 8} + a = 2 \dots \left[\because \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a \right]$$

$$\therefore \frac{\log(2)^5}{\log(2)^3} + a = 2$$

$$\therefore \frac{5 \log 2}{3 \log 2} + a = 2$$

$$\therefore \frac{5}{3} + a = 2$$

$$\therefore a = 2 - \frac{5}{3}$$

$$\therefore a = \frac{1}{3}$$

$$\therefore a = \frac{1}{3} \text{ and } b = \frac{3}{2}$$